LOG-CONCAVITY OF ALL DERIVATIVES OF THE JACOBI-THETA KERNEL

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ABSTRACT. This note records a statement refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by rigorous interval arithmetic.

Summary: Coffey–Csordas Conjecture 2.5 / Csordas Problem 4.13 (2014) *Certificate:* Rigorous interval evaluation of $J_9(1/50) < 0$.

Posed by Coffey and Csordas (2014); counterexample found by GPT-5.5 (Pro) (2026-02-22); formalized by S. Sra; audited by S. Sra.

Write $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ for the Riemann ξ-function and $\Xi(x) = \xi(\frac{1}{2} + ix)$ for its restriction to the critical line, a real even entire function of order one. Pólya's Fourier-cosine representation of Ξ (see Csordas, 2014, §4, who writes $H(x)$ for $\Xi(x/2)/8$) is

$$\Xi(x/2) = 8\int_0^\infty \Phi(t)\cos(xt)\,dt,$$

where Φ is the kernel displayed in the statement. The Riemann Hypothesis is equivalent to all zeros of Ξ being real, that is, to Ξ belonging to the Laguerre–Pólya class of real entire functions that are locally uniform limits of real polynomials with only real zeros. Nothing is known that characterizes, in terms of a kernel, when its cosine transform has only real zeros; the program behind the statement below is to find properties of Φ that constrain the zeros of Ξ.

Classical facts about the kernel (Pólya, Wintner; collected as Theorem 4.1 of Csordas, 2014): Φ is even and positive, analytic in the strip $|\operatorname{Im} z| < \pi/8$, satisfies $\Phi'(t) < 0$ for $t > 0$, and every derivative $\Phi^{(n)}(t)$ decays faster than $\exp(-(\pi-\varepsilon)e^{4t})$ as $t \to \infty$. Because Φ is even, each J_n below is even, so the conjecture is a statement about $t \geq 0$.

The interest in derivatives of Φ comes from the Turán inequalities. The Maclaurin coefficients of Ξ are, up to the normalization, the moments $b_k = \int_0^\infty t^{2k}\Phi(t)\,dt$, since $\Xi(x/2) = 8\sum_{k\geq 0}(-1)^k b_k\, x^{2k}/(2k)!$, and a necessary condition for the Riemann Hypothesis (Pólya–Schur) is

$$b_k^2 - \tfrac{2k-1}{2k+1}\, b_{k-1}b_{k+1} \geq 0 \qquad (k \geq 1).$$

These were proved by Csordas, Norfolk and Varga (1986) and by Csordas and Varga (1988) from concavity properties of Φ, in particular the strict concavity of $\log\Phi(\sqrt{t}\,)$ on $t > 0$, which is stronger than the log-concavity of Φ itself. Coffey and Csordas (2013) gave a new proof that Φ is log-concave, the case $n = 1$ below, and conjectured that every derivative of Φ is strictly log-concave as well; Csordas restated that conjecture as Open Problem 4.13.

A positive function g is *strictly log-concave* when $(\log g)'' < 0$, equivalently $(g')^2 - gg'' > 0$. For $n \geq 2$ the derivative $\Phi^{(n-1)}$ need not be positive (already $\Phi' \leq 0$ on $t \geq 0$), and the source takes the inequality $J_n(t) > 0$ itself as the meaning of strict log-concavity of $\Phi^{(n-1)}$; that inequality is what is refuted below.

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From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

Problem 1 (As posed: G. Csordas (2014), Open Problem 4.13, restating Coffey–Csordas Conjecture 2.5)**.** ([5, Conjecture 2.5]) Show that the derivatives of the Jacobi theta function, $\Phi(t)$, are (strictly) log-concave on $\mathbb{R}$; that is, for each $n \in \mathbb{N}$,

$$J_n(t) := (\Phi^{(n)}(t))^2 - \Phi^{(n-1)}(t)\Phi^{(n+1)}(t) > 0 \quad \text{for } t \in \mathbb{R}.$$

Here, by (4.2) of the same paper, the Jacobi theta function (without the usual factor 4) is $\Phi(t) := \sum_{n\geq 1} \pi n^2 \left(2\pi n^2 e^{4t} - 3\right) \exp\left(5t - \pi n^2 e^{4t}\right)$.

Theorem 2 ([**Disproof**]; computer-assisted proof)**.** *At $n = 9$ and $t = 1/50$, one has $J_9(t) < 0$.*

Proof. Put $u = \pi m^2 e^{4t}$ and define integer polynomials

$$P_0(u) = 2u - 3, \qquad P_{k+1}(u) = 4uP_k'(u) + (5 - 4u)P_k(u).$$

Termwise differentiation gives

$$\Phi^{(k)}(t) = \sum_{m\geq 1} \pi m^2 e^{5t-u} P_k(u).$$

Outward-rounded interval summation, with an explicit geometrically bounded tail, yields

$$\Phi^{(8)}(1/50) \approx 2.95940368409056694 \cdot 10^8,$$
$$\Phi^{(9)}(1/50) \approx 5.80442287530267756 \cdot 10^9,$$
$$\Phi^{(10)}(1/50) \approx 2.02040519425950258 \cdot 10^{11},$$

and an interval for $J_9(1/50)$ contained strictly in the negative half-line, centered at

$$-2.61006208371358922 \cdot 10^{19}.$$

The source package contains a high-precision evaluator and a Sage `RealIntervalField` certificate. The negative margin is many orders of magnitude larger than the certified truncation error. □

Remark 3. GPT-5.5 managed to prove the Turán inequalities for $\Phi^{(n)}$ for $n = 2$ and $n = 3$; while we were attempting an inductive proof to generalize those proofs to all n is when the counterexample popped up. However, we have not yet verified whether the Turán inequalities hold for all $n \leq 8$.

Code and certificate

The case is the directory `counterexamples/theta-derivative-log-concavity/` of the archive, at commit `8fa4613`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. Independent Sage interval cross-checks: `verify_theta.sage`. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/theta-derivative-log-concavity
python verify.py
```

- Theorem 2 (result `theta-derivative-log-concavity`): rigorous interval arithmetic.
- `artifacts/certificate.json`: High-precision value of $J_9(1/50)$; the Sage script provides the outward-rounded interval certificate.

AI USAGE

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexample here was found by GPT-5.5 (Pro) (2026-02-22). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

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